

Parallel Processing

Thoai Nam

High Performance Computing Lab (HPC Lab)

Faculty of Computer Science and Engineering

HCMC University of Technology



What we are doing?



Internet of Things (IoT)

(Timothy Chou)

Do

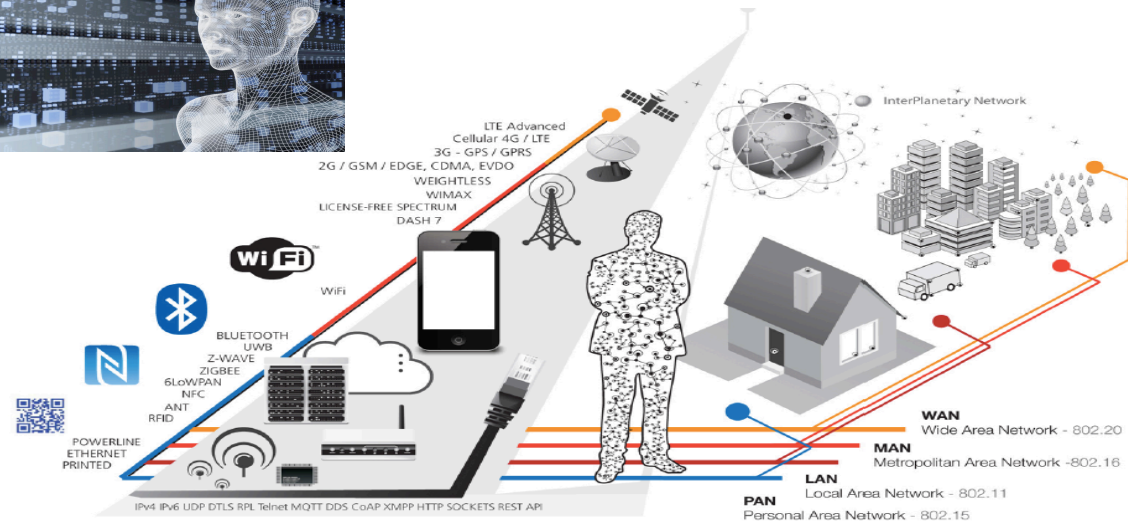


Learn



Collect

Connect



Things



Ambient Light



Touch Screen



Proximity



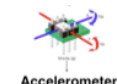
Fingerprint



Attitude



Barometer



Accelerometer



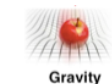
Gyroscope



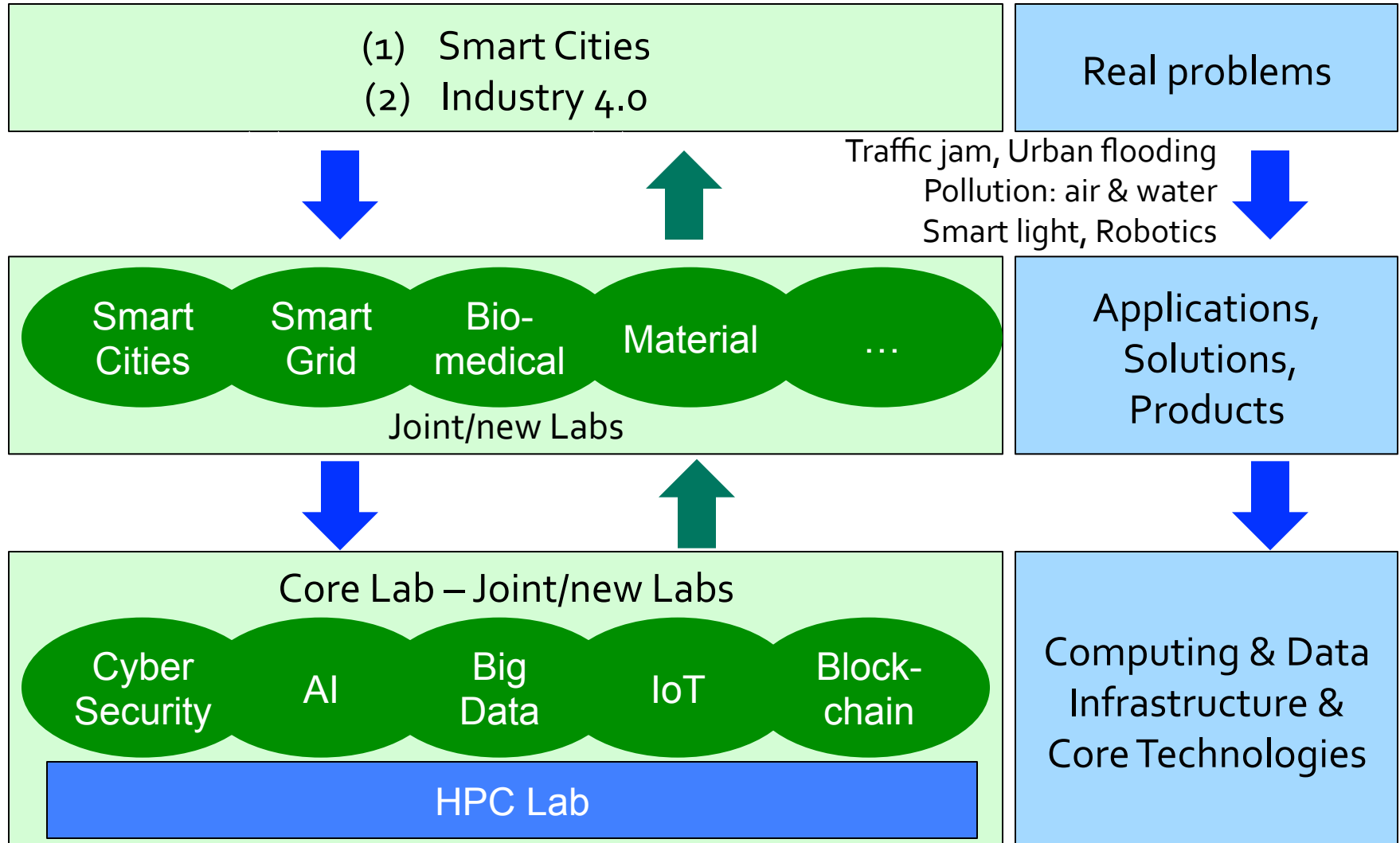
Moisture



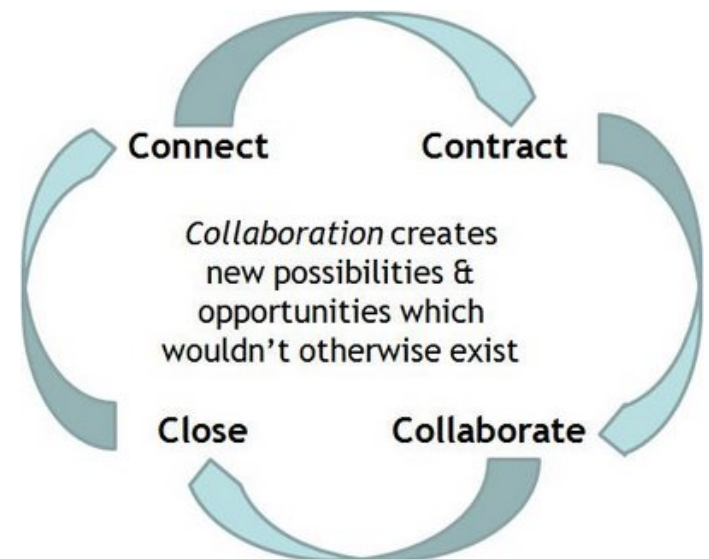
Magnetometer



Gravity



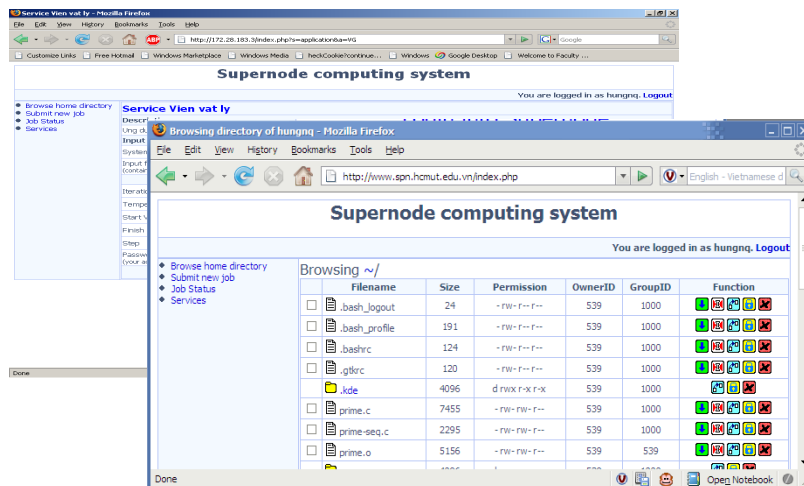
- **Partners**
 - **VNU-HCM (HCMUT)**
 - **Intel**
 - Ho Chi Minh city
- **Plan: 2012-2022**
 - HPC Lab: set up in 2013
 - Strengthen HPC in Vietnam
 - Solving big problems
 - Leading in technology
 - **Partners: HPE (2015) & Nvidia (2017)**
- **Applications: 2013-2022**
 - **Traffic analysis**
 - **Urban flooding**
 - **Big data analytics**



SuperNode I & II



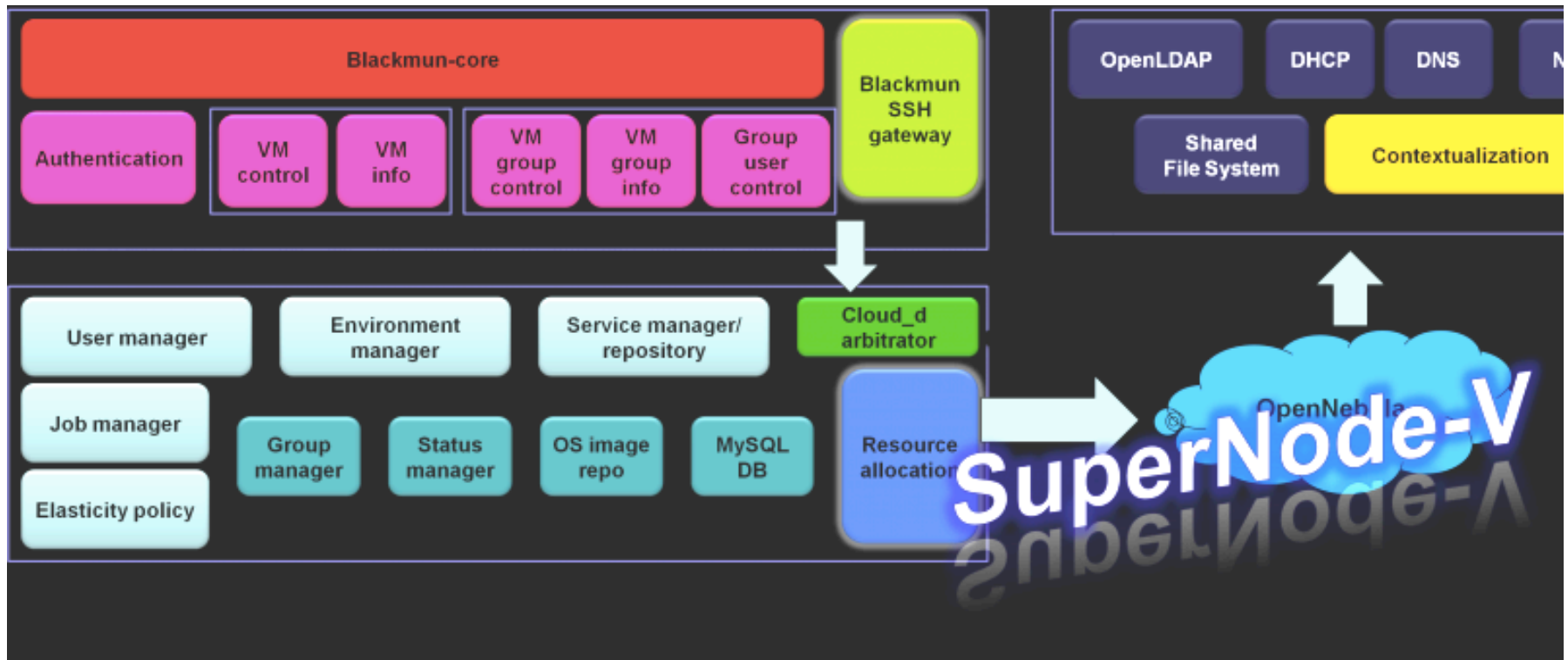
SuperNode I in 1998-2000



SuperNode II in 2003-2005



SuperNode V



SuperNode-V project: 2010-2012

EDA-Grid & VN-Grid

SuperNode II



Applications
Chip design
Data mining
Airfoild optimization

Security

Monitoring



Scheduling

User Management

Campus/VN-Grid (GT)

Resource Management

Information Service

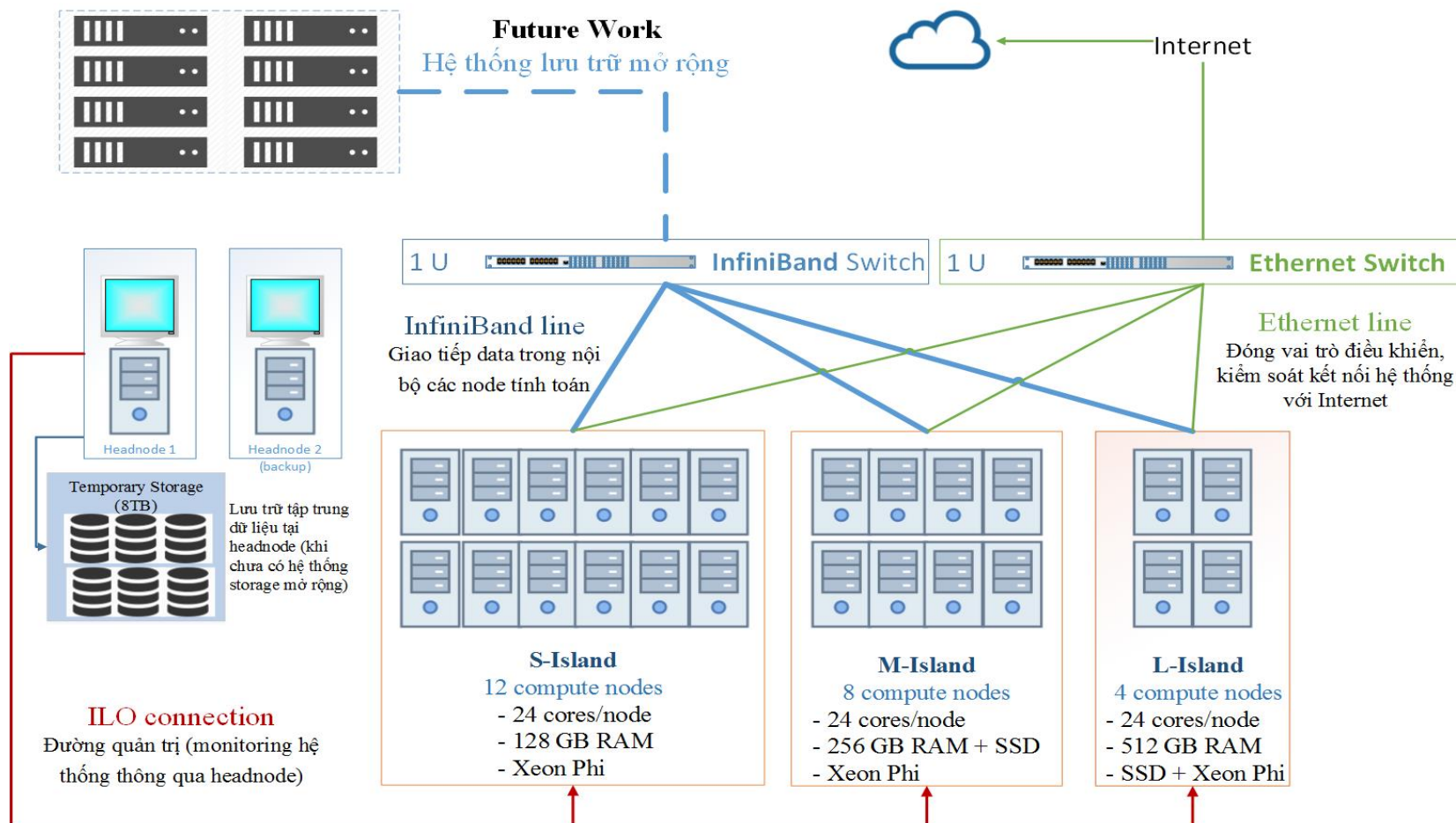
Data Service

POP-C++



SuperNode-XP

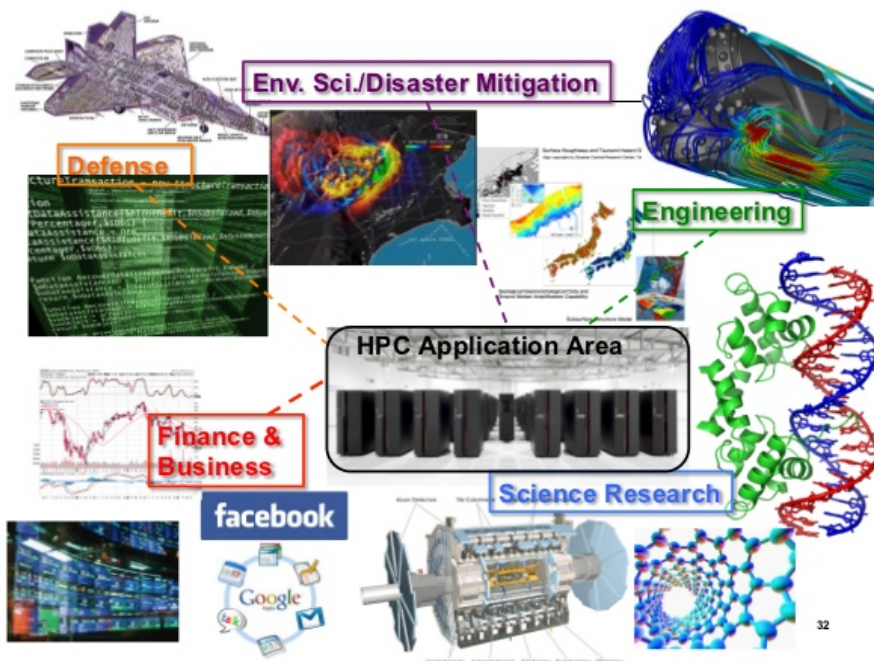
50 TFlops machine



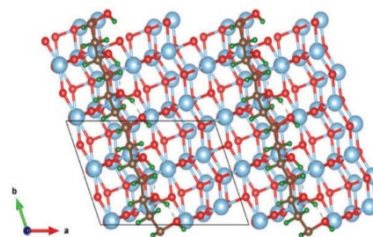
Applications on SuperNode-XP

User	Start Time	End Time	Apps	Organization/Unit
A/Prof. Nguyễn Thống	30/09/14		OpenTelemac	HCMUT – Faculty of Civil Engineering
Ms. Phạm Ngọc Thanh, Dr. Ing Jorg Franke	06/06/16		OpenFoam	Vietnam Germany University
Dr. Lê Thanh Vân	20/08/16		Hadoop	HCMUT – Faculty of Computer Science
A/Prof. Trần Văn Hoài	25/10/16		BLAS	HCMUT – Faculty of Computer Science
Mr. Quân	01/11/16		DFFT	HCMUT – Faculty of Electrical and Electronic Engineering

Ng
A/P
Ho:
Phl
Pro
Dr.
Dr.
Dr.
Dr.
Loi
Dr.

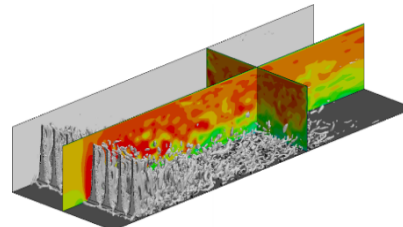


INOMAR center

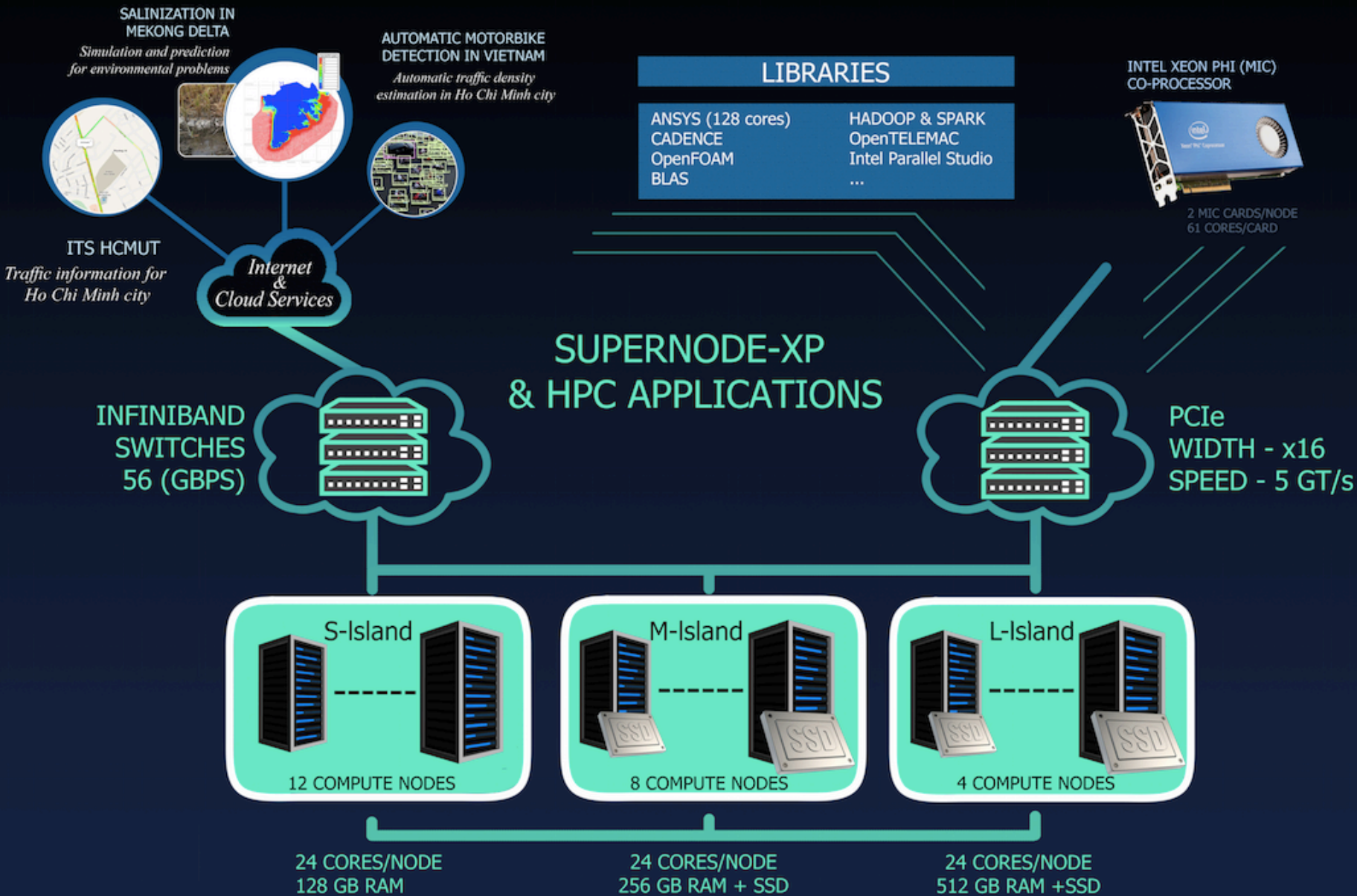


VASP, Quantum
Expresso Simulations

Vietnam Germany University

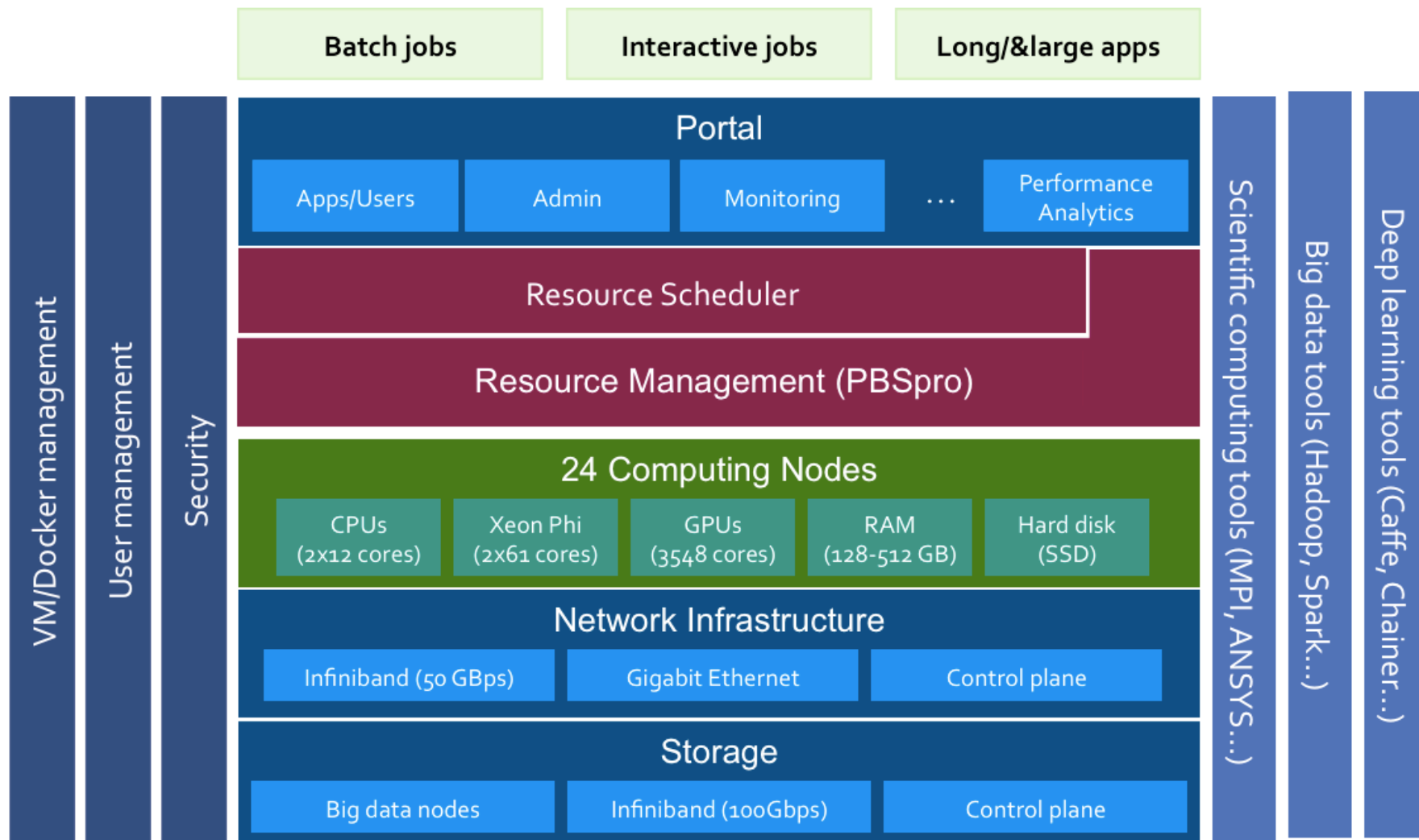


OpenFOAM sims

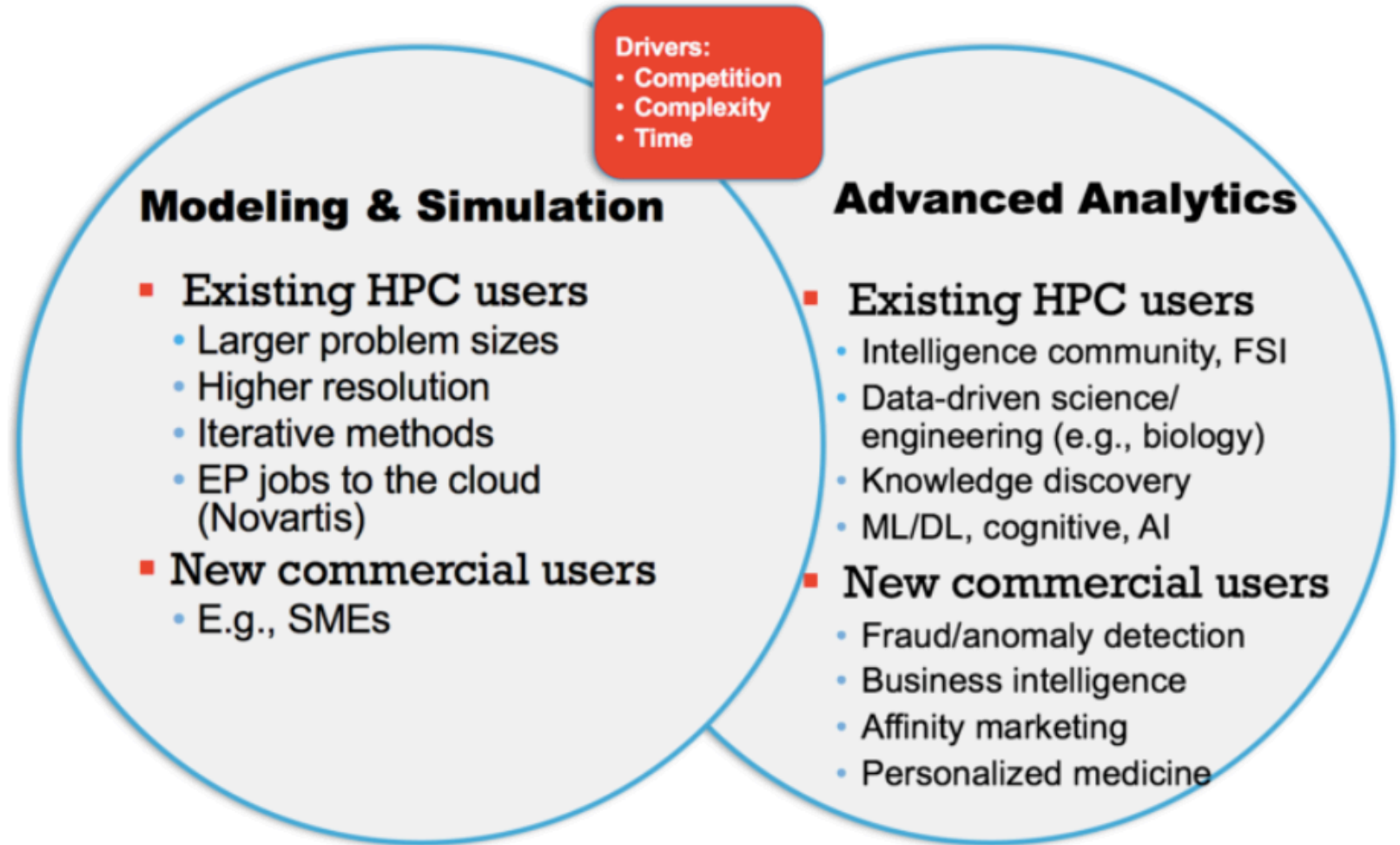




SuperNode-XP Architecture



HPDA = Data-Intensive Computing Using HPC



<https://www.hpcwire.com/2017/04/20/hyperion-idc-paints-bullish-picture-hpc-future/>



- **HPDA (HPC + Big data + AI) platform:**
 - Data collection: data streaming
 - Data analytics: Data mining & AI algorithms
 - Performance
- **Resource management**
 - PBSpro, Yarn + AI scheduling algorithms
- **Networking**
 - MPI One-sided communication
- **Virtualization**
 - Docker & VM
- **Portal & Monitoring**
 - HPC Open OnDemand: <https://openondemand.org>
 - Zabbix: <https://github.com/zabbix/zabbix>
 - Prometheus: <https://prometheus.io>
 - Grafana: <https://grafana.com>
- **Machines**
 - SuperNode-XP/AI
 - 256/512 GB RAM
 - InfiniBand 50, 100, 200 Gbps
 - Storage > 200 TB
- **Tools**
 - MPI, OpenMP
 - Hadoop, Spark, Kafka
 - TensorFlow, Torch, Caffe2
 - PBSpro, Torque
- **Products**
 - SuperNode machine series
 - SuperNode-DataCore: Hybrid Data Platform for HPDA
 - Smart Village/City Data Core



Platform

Mạng



Sử dụng

- Truy cập
- Dữ liệu/kết quả

Hiệu suất

- Multi-core & Many-core (XP, GPU)
- Mạng tốc độ siêu cao: InfiniBand
- Storage (NFS, Lustre, ...)
- HPC/Big data/AI: MPI, Hadoop, Spark, DL
- Kỹ thuật ảo hoá
- Quản lý & phân bổ tài nguyên
- Giải pháp tự động/bán tự động

Quản trị

- Hệ thống
- Người dùng
- Ứng dụng

